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Tutorial -1

This tutorial will test your knowledge of basic C concepts such as variable types, conditions, and loops.

1. Write a program to print Algorithms and Concurrency | 5CS046. Explain its structure.

Ans :

Source Code :

```c
// 1.Write a program to print Algorithms and Concurrency | 5CS046.

#include <stdio.h>
void main(){
  printf("Algorithms and Concurrency | 5CS046");
}

```

Output:

```
praju-maharjan@LOCKON:~/Algorithms_and_Concurrency/Tutorial_1$ ./Q1
Algorithms and Concurrency | 5CS046praju-maharjan@LOCKON:~/Algorithms_and_Co
ncurrency/Tutorial_1$
```

2. Write a program to add two numbers statically (static values assigned to variables) and dynamically (number is input from the users).

Ans:

Source Code :

```c
//2.Write a program to add two numbers statically (static values assigned to variables ) and dynamically ( number is input from the users).

#include <stdio.h>
void main(){
 //Static addition
 int a=1,b=2;
 printf("%d + %d = %d",a,b,a+b);

 //Dynamic addition
 int c,d;
 printf("Enter the first number : ");
 scanf("%d",&c);
 printf("Enter the second number : ");
 scanf("%d",&d);
 printf("%d + %d = %d",c,d,c+d);
}
```

Output:

```
praju-maharjan@LOCKON:~/Algorithms_and_Concurrency/Tutorial_1$ ./Q2
1 + 2 = 3Enter the first number : 5
Enter the second number : 3
5 + 3 = 8praju-maharjan@LOCKON:~/Algorithms_and_Concurrency/Tutorial_1$
```

3. Write a program to print the simple interest with values p, t, and r given by the user, where p is an integer, t is an integer, r is a float, and the interest is a float.

Ans:

Source Code :

```c

#include<stdio.h>
void main(){
 int p,t;
 float r,interest;

 printf("Enter principal amount : ");
 scanf("%d",&p);
 printf("\nEnter time : ");
 scanf("%d",&t);
 printf("\nEnter rate of interest : ");
 scanf("%f",&r);

 interest=(p*t*r)/100;

 printf("Simple interest = %f",interest);
}
```

Output:

```
Enter principal amount : 200

Enter time : 2

Enter rate of interest : 10
Simple interest = 40.000000pr
```

4. Write a program to convert degree celsius to fahrenheit and degree fahrenheit to degree celsius.

Ans:

Source Code:

```c
#include<stdio.h>
#include<stdbool.h>

void main(){
 char choice;
 float temp;

 while(true){
 printf("What do you want to convert : \n1.Celsius to Fahrenheit\n2.Fahrenheit to Celsius\n");
 scanf(" %c",&choice);
 if(choice=='1'){
  printf("Enter temperature in celsius : ");
  scanf("%f",&temp);
  printf("%f° Celsius = %f° Fahrenheit\n",temp,temp*1.8+32);
 }else if(choice=='2'){
  printf("Enter temperature in fahrenheit : ");
  scanf("%f",&temp);
  printf(" \n%f° Fahrenheit = %f° Celsius\n",temp,(temp-32)/1.8);
 }else{
  printf("Please select valid option\n");
  continue;
 }
 printf("Press s to stop. Press any other character to continue.\n");
 scanf(" %c",&choice);
 if(choice=='s'){
  break;
 }
 }
 printf("\n");
}
```

Output:

```
What do you want to convert :
1.Celsius to Fahrenheit
2.Fahrenheit to Celsius
5
Please select valid option
What do you want to convert :
1.Celsius to Fahrenheit
2.Fahrenheit to Celsius
1
Enter temperature in celsius : 100
100.000000° Celsius = 212.000000° Fahrenheit
Press s to stop. Press any other character to continue.
h
What do you want to convert :
1.Celsius to Fahrenheit
2.Fahrenheit to Celsius
2
Enter temperature in fahrenheit : 32

32.000000° Fahrenheit = 0.000000° Celsius
Press s to stop. Press any other character to continue.
s
```

5. Explain operators and their types in C. *https://www.w3schools.com/c/c_operators.php*

Ans:

In C programming, an operator is a symbol that performs a specific operation on one or more operands. Operators are the building blocks of C expressions and can be classified into several types. Arithmetic operators like +, -, *, /, and % are used for basic mathematical calculations. Relational operators such as ==, !=, >, <, >=, and <= compare values and return true or false. Logical operators like &&, ||, and ! are used to combine or negate conditions, while assignment operators (=, +=, -=, *=, /=, %=) are used to assign or update variable values. Increment (++) and decrement (--) operators increase or decrease a variable by one, and bitwise operators (&, |, ^, ~, <<, >>) manipulate data at the binary level. C also has the conditional (ternary) operator ?:, which acts as a shorthand for if-else statements, and special operators like sizeof, ,, and pointer operators * and &. Understanding these operators and how they interact is crucial for writing efficient and correct C programs.

6. Write a program to print whether the value is an odd number or an even number.

Ans:

Source Code:

```c

#include<stdio.h>
void main(){
 int num[2];
 printf("Enter two numbers : ");
 scanf("%d%d",&num[0],&num[1]);

 for(int i=0;i<2;i++){
  if(num[i]%2==0){
   printf("%d is an even number.\n",num[i]);
  }else{
   printf("%d is an odd number.\n",num[i]);
  }
 }
printf("\n");
}
```

Output:

```
Enter two numbers : 32546
1435211
32546 is an even number.
1435211 is an odd number.
```

7. Write a program to find the greatest of two numbers and the greatest of three numbers.

Ans:

Source Code:

```c
#include<stdio.h>
void main(){
 int g2_1,g2_2,g3_1,g3_2,g3_3;

 printf("Enter two numbers : ");
 scanf("%d%d",&g2_1,&g2_2);
 if(g2_1>g2_2){
  printf("%d is greater.\n",g2_1);
 }else{
  printf("%d is greater.\n",g2_2);
 }

 printf("Enter three numbers : ");
 scanf("%d%d%d",&g3_1,&g3_2,&g3_3);
 if(g3_1>g3_2 && g3_1>g3_3){
  printf("%d is greatest.",g3_1);
 }else if(g3_2>g3_1 && g3_2>g3_3){
  printf("%d is greatest.",g3_2);
 }else{
  printf("%d is greatest.",g3_3);
 }
 printf("\n");
}
```

Output:

```
Q7
Enter two numbers : 2
5
5 is greater.
Enter three numbers : 1321
41325
12532
41325 is greatest.
```

8. Explain the switch statement with syntax, and provide an example on switch statement.
   https://www.w3schools.com/c/c_switch.php

Ans:

In C, the switch statement is a control structure used to execute one block of code from multiple options based on the value of an expression, usually an integer, char, or enum. It compares the expression against different case labels, and when a match is found, the corresponding block of code runs. Each case is typically followed by a break statement to stop execution from falling through to the next case. An optional default case can be included to handle values that do not match any of the cases. This makes the code cleaner and easier to read compared to multiple if-else statements.

Syntax:

```
switch(expression) {
   case constant1:
      // code to execute if expression == constant1
      break;
   case constant2:
      // code to execute if expression == constant2
      break;
   ...
   default:
      // code to execute if no case matches
}
```

Example:

```c

#include <stdio.h>

void main(){
 int day_num;
 printf("Enter a number from 1 to 7 : ");
 scanf("%d",&day_num);
 switch(day_num){
  case 1:
   printf("Sunday\n");
   break;
  case 2:
   printf("Monday\n");
   break;
  case 3:
   printf("Tuesday\n");
   break;
  case 4:
   printf("Wednesday\n");
   break;
  case 5:
   printf("Thursday\n");
   break;
  case 6:
   printf("Friday\n");
   break;
  case 7:
   printf("Saturday\n");
   break;
   default:
   printf("Invalid input\n");
 }
}
```

Output:

```
Q7
Enter a number from 1 to 7 : 4
Wednesday
```

9. Write a program to print the odd and even numbers from 1 to 100.

Ans:

Source Code:

```
#include<stdio.h>
void main(){
 printf("Odd numbers\tEven numbers\n");
 for(int i=1;i<=100;i++){
  if(i%2==0){
  printf("\t\t\t%d \n",i);
  }else{
  printf("%d",i);
  }
 }
 printf("\n");
}
```

Output:

```
Odd numbers     Even numbers
1                       2
3                       4
5                       6
7                       8
9                       10
11                      12
13                      14
15                      16
17                      18
19                      20
21                      22
23                      24
25                      26
27                      28
29                      30
31                      32
33                      34
35                      36
37                      38
39                      40
41                      42
43                      44
45                      46
47                      48
49                      50
51                      52
53                      54
55                      56
57                      58
59                      60
61                      62
63                      64
65                      66
67                      68
69                      70
71                      72
73                      74
75                      76
77                      78
79                      80
81                      82
83                      84
85                      86
87                      88
89                      90
91                      92
93                      94
95                      96
97                      98
99                      100
```

10. Write a program to print the multiplication table of the user-given number.
Ans:

Source Code:

```c
#include<stdio.h>
void main(){
 int num;
 printf("Enter a number : ");
 scanf("%d",&num);
 printf("Multiplication table of %d\n",num);
 for(int i=1;i<=10;i++){
  printf("%d x %d = %d.\n",num,i,num*i);
 }
 printf("\n");
}
```

Output:

```
Enter a number : 5
Multiplication table of 5
5 x 1 = 5.
5 x 2 = 10.
5 x 3 = 15.
5 x 4 = 20.
5 x 5 = 25.
5 x 6 = 30.
5 x 7 = 35.
5 x 8 = 40.
5 x 9 = 45.
5 x 10 = 50.
```

The End